

Prediction Accuracy Improvement for Cardiovascular Diseases Using Machine Learning Algorithm

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Abstract— Worldwide Cardiovascular diseases (CVDs) are one of the most common causes of death. Multiple ML algorithms are employed in this work to address the CVD problem. The technique used in this study employs several machine-learning algorithms to predict CVD. To predict cardiac sickness, data mining approaches such as Decision Trees (DT), Naive Bayes (NB), Support Vector Machine (SVM), Logistic Regression (LR), Random Forest (RF), Ada Boosting, Gradient Boosting, and Bagging algorithms have been utilized. Using Base learners and Ensemble classifiers, this study tried to predict the presence of heart disease. Bagging and Gradient Boosting Classifiers had the highest accuracy, with 81.3% and 81.3%, respectively.

Keywords— Cardiovascular diseases, Machine learning algorithm, Classifiers, Bagging, Boosting.

I. INTRODUCTION

Cardiovascular diseases (CVDs) are one of the top causes of death globally. According to estimates, 17.9 million people died from CVDs in 2019, accounting for 32% of all fatalities worldwide. Coronary artery disease and strokes were responsible for 85% of these fatalities [1]. Moreover, poor and middle-income nations account for more than three-quarters of CVD mortality. CVDs accounted for 38% of the 17 million unexpected deaths (aged 70 and under) caused by noncommunicable illnesses in 2019 [2]. Most cardiovascular illnesses may be prevented by eliminating risk factors such as cigarette smoking, poor nutrition and weight management, physical inactivity, and harmful alcohol use. It is critical to recognize cardiovascular disease as soon as possible so that executives with guidance and medications can begin [3].

Poor eating habits, physical inactivity, cigarette use, and problematic alcohol use are the most significant societal risk factors for cardiovascular disease and stroke. Humans may have a higher circulatory strain, increased blood glucose, increased blood lipids, and obesity and weight as a result of social hazard factors [4]. At critical assessment workplaces, these "halfway risk factors" may be computed and show a higher risk of coronary episode, stroke, cardiovascular breakdown, and other problems [5]. Cigarette cessation, dietary

salt reduction, consumption of more locally produced foods, regular physical exercise, and abstinence from dangerous alcohol intake have all been demonstrated to lessen the risk of cardiovascular disease. Wellness approaches that provide favorable circumstances for making sound decisions in a fair and accessible way are crucial for persuading individuals to embrace and maintain healthy activities [6].

Developing endurance for cardiac breakdowns (and CVDs in general) is still a challenge these days, both as far as accomplishing high-expectation precision and distinguishing the driving elements. The bulk of models developed for this purpose achieve only inconspicuous exactness, with little interpretability from the anticipated elements. Subsequent models demonstrate improvements, especially when the endurance result is paired with added objectives such as hospitalization. Although researchers have distinguished an expansive arrangement of indicators and markers, there is no common agreement on their general effect on endurance forecast. As called attention to by Sakamoto and partners, the present circumstance is generally because of an absence of reproducibility, which forestalls reaching authoritative determinations about the significance of the recognized components. Further, this absence of reproducibility emphatically influences model exhibitions: speculation about outside approval datasets is regularly conflicting and accomplishes just unassuming segregation. Thus, hazard scores refined from the models experience comparable issues, restricting their unwavering quality. Such a vulnerability has prompted the expansion of new danger scores showing up in the writing somewhat recently, with blended outcomes. As a halfway answer for working on models' adequacy, as of late distributed examinations included associates confined to explicit classes of patients (for example, diabetic). These efforts have resulted in conventional models and risk indexes with better but not optimum implementation. The forecasting approach's goal is to design a framework that can infer typical for an anticipated class from a mix of varied facts.

In addition, there are other basic factors of CVDs. These are the major forces causing social, economic, and societal transformation - globalization, urbanization, and population

maturation. Insecurity, anxiety, and inherent factors are some of the different measures of CVDs. Furthermore, pharmacological therapy for high blood pressure, diabetes, and high blood fat is critical to lowering cardiovascular risk and preventing pulmonary difficulties or strokes in people with these illnesses. The paper's organization is as follows; section 2 presents the proposed methodology, and section 3 discusses the results. Finally, section 4 discusses the main conclusions and future work.

II. LITERATURE REVIEW

AI applied to clinical records, specifically, can be a viable instrument both to foresee the endurance of every tolerant having cardiovascular breakdown manifestations and to recognize the main clinical provisions (or hazard factors) that might prompt cardiovascular breakdown. Researchers can exploit AI for the clinical forecast as well as for highlight positioning [7]. Whenever applied to clinical data or coupled with imaging, computerized learning demonstrates its predictive power. Furthermore, deep learning and meta-investigation ideas applied to this sector have recently appeared in the literature, increasing human specialists' performance [8]. C4.5-based classification has been used to improve heart disease prediction. This heart disease identification system achieves 89% accuracy. Multilayer's Perceptron Neural Network with Back-Propagation (MLPNN-BP) technique has been employed to improve heart disease prediction accuracy [7]. This study gives improved results with an accuracy of 98.58 %. Different classification techniques have been used. This paper proposed an effective Heart Disease Prediction System (EHDPS) model and gave improved results with an accuracy of 87.5% [9]. Four classification algorithms (Naïve Bayes, K-NN, DT, RF) have been used. According to this research, the K-NN Method is the most accurate method, with 90.789% accuracy [10]. Seven classifiers namely K-NN, Naive Bayes, J48, J-Rip, SVM, Ada-boost, SGD and DT classifiers, have been used to improve heart disease prediction accuracy. According to this research, the K-NN Method is the most accurate method, with 90.7073% accuracy [11]. A comprehensive review of different data mining techniques has been done and results show that the maximum number of researchers used the heart disease dataset which comprises only 303 cases with 14 features. Which is small and restrictive. To obtain a more generalized classification, new heart disease datasets with more information from geographically different sources are required [12]. Convolutional neural network algorithm used to improve heart disease accuracy. By using this model accuracy goes up to (85-87%) [13].

This research demonstrates that feature selection strategies are effective. The RF ensemble approach combines correlation and LR to get superior outcomes with 99% accuracy. On a dataset containing OH-encoded categorical characteristics, optimized XGBoost was applied. This study indicates that using this model accuracy scores of 91.80% are achieved in predicting heart disease. Cardiovascular disease is frequently predicted with accuracy. Additional input variables, such as controllable and uncontrolled risk factors, might lead to more accurate findings. Researchers present a classification technique that combines Naive Bayes and genetic algorithms

to forecast a patient's cardiac status in this research. Throughout this publication, a neural network is used to create a heart condition prediction system (HDPS) [14]. The HDPS system predicts if a patient will develop a cardiac issue. The technology makes predictions based on 13 medical indicators such as genre, blood pressure, and cholesterol. Incorporating machine learning approaches such as ANN and MLPNN, the EHDPS model was proposed for efficient heart disease diagnosis. Using neural networks, the system predicts heart illness with 100% accuracy [15].

A hybrid intelligent system framework for heart disease prediction was developed employing seven well-known classifiers, including logistic regression, K-NN, ANN, SVM, NB, DT and random forest [16-19]. According to the data, the classifier logistic regression with 10-fold cross-validation has the highest accuracy of 89% [20]. A three-step feature selection-based approach for data mining in HD diagnosis has been developed. CFS and LLCFS had the best outcomes. A hybrid technique based on CSA and COA filter-based feature selection is suggested. According to the results, the CSA filter-based FS beat the COA filter-based FS. With a 94.00% accuracy. Proposed CFNN-PCA-based hybrid system to improve HDP. The accuracy rate of principal component analysis (PCA) improved to 97.7%, according to the findings. Using different data mining approaches, create a system for predicting cardiac disease. According to this research, the proposed approach has the highest accuracy of 97.14%. For the prediction of cardiac illness, a hybrid model (NB+PSO) is proposed. The accuracy of this model is 87.91%. Develop optimized-DBN-based HDPS. Optimized DBN shows the best result of 82.94% accuracy [21].

III. METHODOLOGY

A. Heart Disease Dataset

Kaggle was used to obtain the cardiovascular disease datasets in Figure 1. There are seventy-six qualities in this data; nonetheless, all disclosed experiments consult with a set of 14 of them. Its complete range of 303 rows and 14 columns among that one hundred sixty-five have heart disease.

```
#read the csv dataset
data = pd.read_csv("file:/C:/Users/hp/Desktop/jamshaid123/heart.csv", encoding='ANSI')
data.columns
data.head()
```

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	target
0	63	1	3	145	233	1	0	150	0	2.3	0	0	1	1
1	37	1	2	130	250	0	1	187	0	3.5	0	0	2	1
2	41	0	1	130	204	0	0	172	0	1.4	2	0	2	1
3	56	1	1	120	236	0	1	178	0	0.8	2	0	2	1
4	57	0	0	120	354	0	1	163	1	0.6	2	0	2	1

Fig. 1. Dataset diagram.

B. Data Preprocessing

Data preprocessing includes working raw records to well-fashioned statistics units so that statistics mining analytics may be applied. Raw statistics are frequently incomplete and have inconsistent formatting. The adequacy or inadequacy of statistics education has an instantaneous correlation with the fulfilment of any task that contains statistics analytics. Preprocessing includes each statistics validation and statistics

imputation. The purpose of statistics validation is to evaluate whether or not the statistics in the query are whole and accurate. The purpose of statistics imputation is to accurately make mistakes and enter lacking values -- both manually or routinely through commercial enterprise system automation (BPA) programming. Data preprocessing is utilized in each database-pushed and rules-primarily based totally application. In gadget mastering (ML) processes, statistics preprocessing is crucial for making sure big datasets are formatted in this sort of manner so that the statistics they incorporate may be interpreted and parsed through mastering algorithms.

C. Feature Selection

Figure 2 shows feature selection isn't the same as dimensionality reduction. Both strategies are seeking for to lessen the variety of attributes inside the dataset, however, a dimensionality reduction technique achieves this by way of means of growing new combos of attributes, in which characteristic choice strategies consist of and exclude attributes gifted with inside their cords without converting them. Irrelevant or partly applicable capabilities can damagingly affect the version routine. To acquire higher accuracy for our version we recognized the surprisingly crucial capabilities of the dataset by forcing a function choice approach.

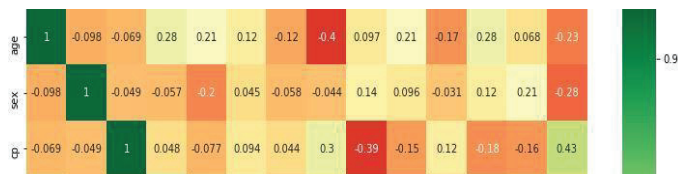


Fig. 2. Selected Feature.

Correlation Matrix with the Heatmap is shown in Figure 3. Correlation states how the capabilities are associated with every different or goal variable. Correlation may be positive (growth in a single cost of function will increase the cost of the goal variable) or negative (growth in a single cost of function decreases the cost of the goal variable).

IV. RESULTS AND DISCUSSIONS

A. Support Vector Machine (SVM)

This work tries to predict the presence of heart ailment by using Base learners and Ensemble classifiers. It was observed that the Accuracy of SVM is shown at 80.2 % in Figure 4.

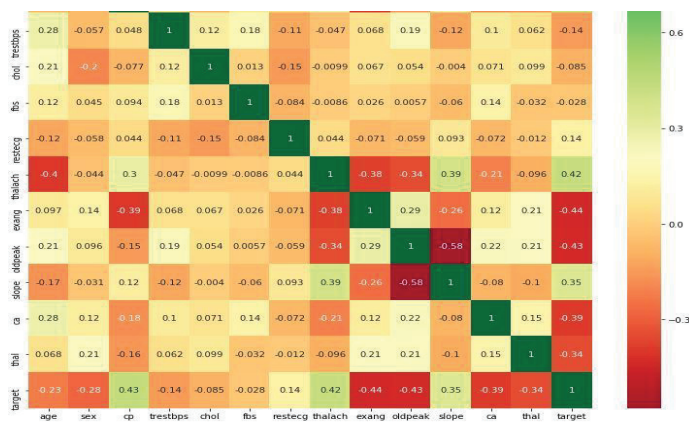


Fig. 3. Heatmap.

SVM accuracy score: 0.8021978021978022

[[37 13]
[5 36]]

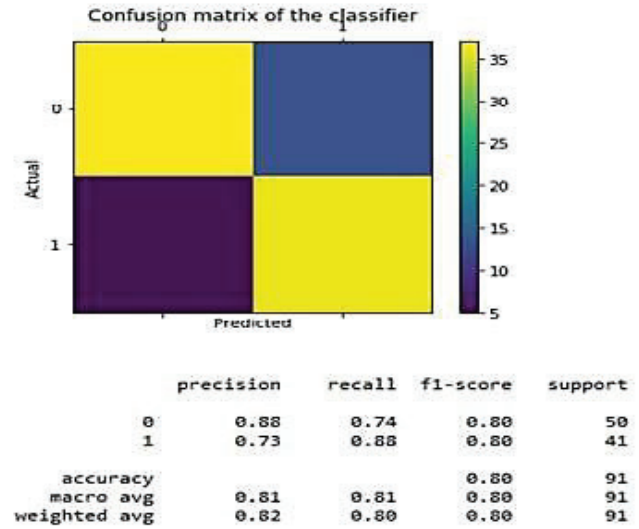


Fig. 4. SVM Result.

B. Naïve Bayes (NB)

This work tries to predict the presence of heart ailment by using Base learners and Ensemble classifiers. It was observed that the Accuracy of Multinomial Naive Bayes is shown at 76.9 % in Figure 5.

C. Logistic Regression (LR)

This work tries to predict the presence of heart ailment by using Base learners and Ensemble classifiers. It was observed that the Accuracy of Logistic Regression is shown at 80.2% in Figure 6.

MultinomialNB LogLoss 0.5154524177119627
MultinomialNB accuracy score: 0.7692307692307693

[[34 16]
[5 36]]

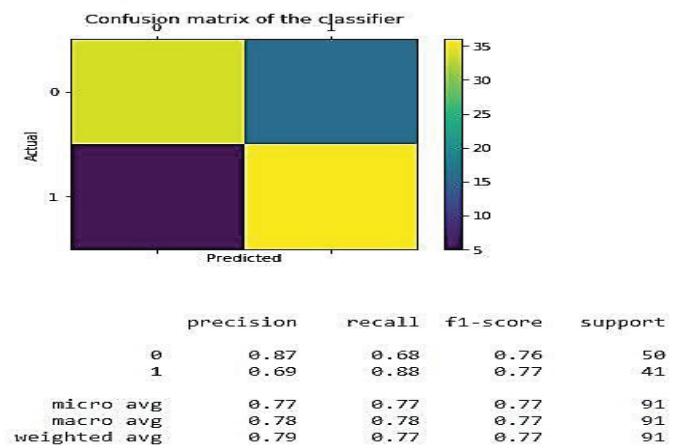
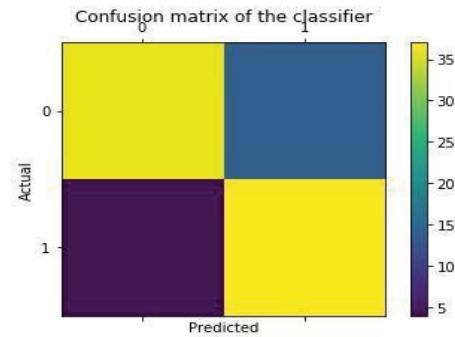


Fig. 5. Naive Bayes Result.

LogisticRegression LogLoss 0.4263220106666439
 Logistic Regression accuracy score: 0.8021978021978022

[[36 14]
 [4 37]]



	precision	recall	f1-score	support
0	0.90	0.72	0.80	50
1	0.73	0.90	0.80	41
micro avg	0.80	0.80	0.80	91
macro avg	0.81	0.81	0.80	91
weighted avg	0.82	0.80	0.80	91

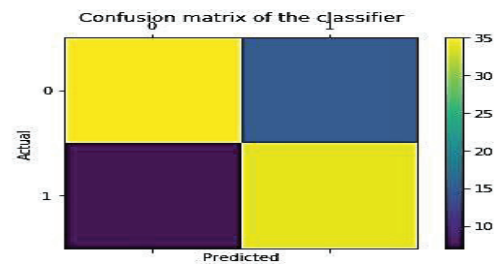
Fig. 6. Logistic Regression Result.

D. Decision Tree (DT)

This work tries to predict the presence of heart ailment by using Base learners and Ensemble classifiers. It was observed that the accuracy of the Decision Tree is shown at 75.8 % in figure 7.

Decision Tree LogLoss 0.8911144637237393
 Decision Tree accuracy score: 0.7582417582417582

[[35 15]
 [7 34]]



	precision	recall	f1-score	support
0	0.83	0.70	0.76	50
1	0.69	0.83	0.76	41
micro avg	0.76	0.76	0.76	91
macro avg	0.76	0.76	0.76	91
weighted avg	0.77	0.76	0.76	91

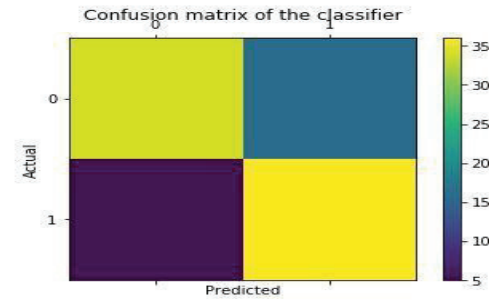
Fig. 7. Decision Tree Result.

E. Random Forest (RF)

This work tries to predict the presence of heart ailment by using Base learners and Ensemble classifiers. It was observed that the Accuracy of Random Forest is shown at 76.9 % in Figure 8.

RandomForest LogLoss 1.1711416214173578
 Random Forest accuracy score: 0.7692307692307693

[[34 16]
 [5 36]]



	precision	recall	f1-score	support
0	0.87	0.68	0.76	50
1	0.69	0.88	0.77	41
micro avg	0.77	0.77	0.77	91
macro avg	0.78	0.78	0.77	91
weighted avg	0.79	0.77	0.77	91

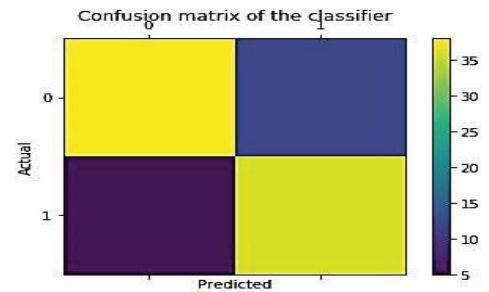
Fig. 8. Random Forest Result.

F. Bagging

This work tries to predict the presence of heart ailment by using Base learners and Ensemble classifiers. It was observed that the Accuracy of Bagging is shown at 81.3 % in Figure 9.

bagging_accuracy score: 0.8131868131868132

[[38 12]
 [5 36]]



	precision	recall	f1-score	support
0	0.88	0.76	0.82	50
1	0.75	0.88	0.81	41
micro avg	0.81	0.81	0.81	91
macro avg	0.82	0.82	0.81	91
weighted avg	0.82	0.81	0.81	91

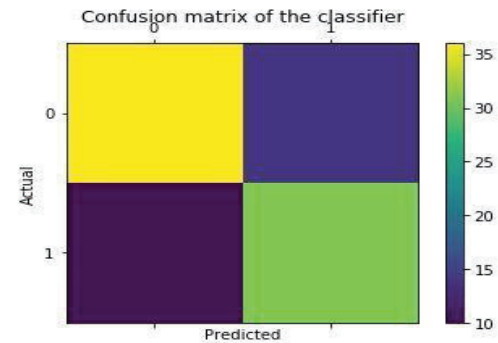
Fig. 9. Bagging Result

G. Adaptive Boosting

This work tries to predict the presence of heart ailment by using Base learners and Ensemble classifiers. It was observed that the Accuracy of Adaptive Boosting is shown at 73.6 % in Figure 10.

Ada_boost_accuracy score: 0.7362637362637363

```
[[36 14]
 [10 31]]
```



	precision	recall	f1-score	support
0	0.78	0.72	0.75	50
1	0.69	0.76	0.72	41
micro avg	0.74	0.74	0.74	91
macro avg	0.74	0.74	0.74	91
weighted avg	0.74	0.74	0.74	91

Fig. 10. Adaptive Boosting Result

H. Gradient Boosting

This work tries to predict the presence of heart ailment by using Base learners and Ensemble classifiers. It was observed that the Accuracy of Gradient Boosting is shown at 81.3% in Figure 11.

Gradient_boosting_accuracy score: 0.8131868131868132

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[[39 11]
 [ 6 35]]
```

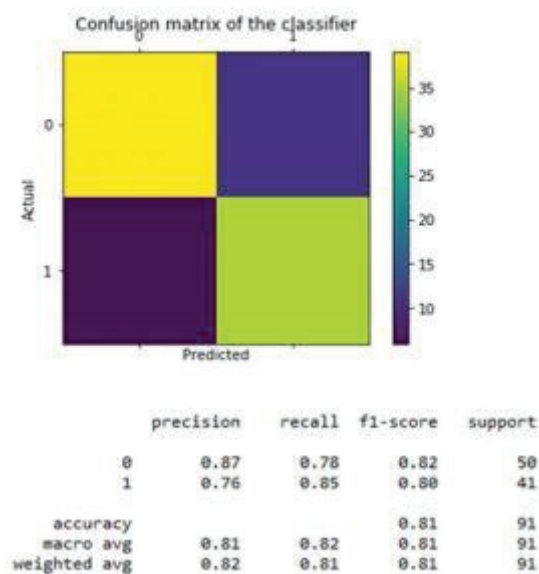


Fig. 11. Gradient Boosting Result

Table 1 Comparative study of different machine learning algorithms (SVM, NB, LR, DT, RF, Bagging, Ada Boosting and Gradient Boosting).

TABLE I. COMPARATIVE STUDY OF DIFFERENT ML ALGORITHMS

No.	Base Learners	Precision	Recall	F1-Score	Accuracy
1	SVM	0.82	0.80	0.80	0.802
2	NB	0.79	0.77	0.77	0.769
3	LR	0.82	0.80	0.80	0.802
4	DT	0.77	0.76	0.76	0.758
5	RF	0.79	0.77	0.77	0.769
6	Bagging	0.82	0.81	0.81	0.813
7	Ada Boosting	0.74	0.74	0.74	0.736
8	Gradient Boosting	0.82	0.82	0.82	0.813

Figure 12 shows that bagging and gradient boosting show better accuracy, precision, recall and F1 scores as compared to other algorithms.

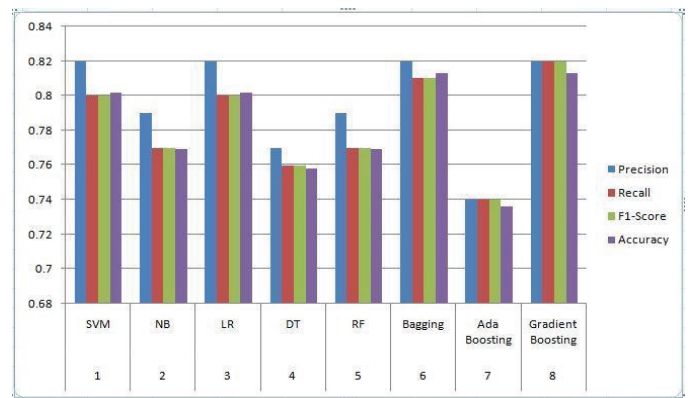


Fig. 12. Comparison Result

V. CONCLUSIONS AND FUTURE WORK

An approach that employs eight machine learning algorithms to anticipate cardiovascular issues is successful. Bagging and Gradient boosting classifiers had the highest accuracy of 81.3% and 81.3%, respectively. Data mining methods such as decision trees, Naive Bayes, SVM, LR, RF, ADA boosting, gradient boosting, and Bagging algorithms have been used to predict cardiac illness. The study attempts to predict the existence of heart disease using Base learners and Ensemble classifiers. The Accuracy of Bagging and Gradient Boosting Classifiers (81.3%) was found to be the highest. To obtain more generalized classification results, multiple heart disease datasets from geographically diverse sources with more features are needed.

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